CONTINUITY

SYNOPSIS

Continuity at a point :

- A function f is said to be continuous at 'a' if f is defined in a neighbourhood of 'a' and $\underset{x \to a}{Lt} f(x) = f(a)$.

  i) If $\underset{x \to a^-}{Lt} f(x) = f(a)$ then $f(x)$ is left continuous at $x = a$.

  ii) If $\underset{x \to a^+}{Lt} f(x) = f(a)$ then $f(x)$ is right continuous at $x = a$.

W.E-1:

The function $f(x) = \dfrac{(3^x - 1)^2}{\sin x \,.\, ln(1+x)}, x \neq 0$ ***is continuous at*** $x = 0$. ***Then the value of*** $f(0)$ ***is***

Sol: Given $f(0) = \lim\limits_{x \to 0} \dfrac{(3^x - 1)^2}{\sin x . ln(1+x)}$

$$\Rightarrow f(0) = \lim_{x \to 0} \frac{\left(\frac{3^x - 1}{x}\right)^2}{\left(\frac{\sin x}{x}\right) . \left(\frac{ln(1+x)}{x}\right)} = (ln\, 3)^2$$

- A function f is said to be continuous in an open interval (a,b) if it is continuous at each and every point in the interval (a,b).
- A function f is said to be continuous on $[a,b]$ if

  (i) f is continuous at each point of (a,b)

  (ii) f is right continuous at $x = a$

  (iii) f is left continuous at $x = b$

W.E-2:

Let f ***be a continuous function on*** $[1,3]$. ***If*** f ***takes only rational values for all*** x ***and*** $f(2) = 10$ ***then*** $f(1.5)$ ***is equal to***

Sol : f(x) is continuous function on [1,3] and takes only rational values then f(x) is constant function.

$\therefore f(2) = f(1.5) = 10$

Discontinuity :

- If $f(x)$ is not continuous at $x = a$, we say that $f(x)$ is discontinuous at $x = a$.

  $f(x)$ will be discontinuous at $x = a$ in any of the following cases :

  i) $\lim\limits_{x \to a^-} f(x)$ and $\lim\limits_{x \to a^+} f(x)$ exist but are not equal.

  (ii) $\lim\limits_{x \to a^-} f(x)$ and $\lim\limits_{x \to a^+} f(x)$ exist and are equal but not equal to $f(a)$

  (iii) $f(a)$ is not defined

  (iv) At least one of the limit doesn't exist.

W.E-3:

Let $f(x)$ ***be defined in the interval*** $[0,4]$ ***such that***

$$f(x) = \begin{cases} 1 - x, & 0 \le x \le 1 \\ x + 2, & 1 < x < 2 \\ 4 - x, & 2 \le x \le 4 \end{cases}$$ ***then number of points where*** $f(f(x))$ ***is discontinuous is***

Sol: $f(x)$ is discontinuous at x = 1 and x = 2

$\Rightarrow f(f(x))$ is discontinuous when f(x) = 1 & 2

Now $1 - x = 1 \Rightarrow x = 0,$ where $f(x)$ is continuous

$x+2=1 \Rightarrow x=-1 \notin (1,2)$

$4-x=1 \Rightarrow x=3 \in [2,4]$

Now, $1-x=2 \Rightarrow x=-1 \notin [0,1]$

$x+2=2 \Rightarrow x=0 \notin (1,2]$

$4-x=2 \Rightarrow x=2 \in [2,4]$

Hence, $f(f(x))$ is discontinuous at two points, x = 2, 3.

- If f and g are continuous functions of x at $x=a$, then the following functions are continuous at $x=a$.

i) $f+g$ ii) $f-g$ iii) $f.g$

iv) cf if $c \in R$ v) $\frac{f}{g}$ if $g(a) \neq 0$.

Note:

Even if f and g are not continuous at $x=a$, $f \pm g, f.g, \frac{f}{g}$, gof may be continuous at $x=a$.

- If f is continuous at $x=a$ and g is continuous at $f(a)$, then (gof) is continuous at $x=a$.
- If f is contiunuous in $[a,b]$ then it is bounded in $[a,b]$. i.e there exist k and m such that $k \le f(x) \le m, \forall x \in [a,b]$ where k and m are minimum and maximum values of $f(x)$ respectively in the interval $[a,b]$.

In this case f takes every real value between k and m at least once. Thus range of f is $[k,m]$

- If f is continuous on $[a,b]$ such that $f(a)$ and $f(b)$ are of opposite signs then there exist at least one solution for the equation $f(x)=0$ in the interval (a,b)

Types of discontinuity :

- **Discontinuity of first kind (or) Removable discontinuity :**

If $\lim_{x \to a} f(x)$ exists but is not equal to $f(a)$ (or) $f(a)$ not difined then the f is said to have a removable discontinuity at $x=a$. It is also called discontinuity of the 1st kind. In this case we can redefine the function by making $\lim_{x \to a} f(x) = f(a)$ and make it continuous at $x=a$.

Removable discontinuities are of two types
1) Missing point discontinuity
2) Isolated point discontinuity

- **Missing point discontinuity :**

$\lim_{x \to a} f(x)$ exists finitely and $f(a)$ is not defined.

Example: $f(x) = \frac{(2-x)(x^2-8)}{(2-x)}$ has a missing point discontinuity at $x=2$.

- **Isolated point discontinuity :**

$\lim_{x \to a} f(x)$ exists finitely and $f(a)$ is defined but $\lim_{x \to a} f(x) \neq f(a)$.

Example: $f(x) = [x]+[-x]$ has isolated point discontinuities at all integers.

- **Discontinuity of second kind (or) irremovable discontinuity :**

A function $f(x)$ is said to have a discontinuity of the second kind at $x=a$ if $\lim_{x \to a} f(x)$ does not exist.

Irremovable discontinuities are of three types
1) Finite discontinuity (or) jump discontinuity
2) Infinite discontinuity
3) Oscillatory discontinuity

- **Finite Discontinuity:**

$\lim_{x \to a^+} f(x)$, $\lim_{x \to a^-} f(x)$ are both finite and are not equal.

Example: $f(x) = \frac{1}{1+2^{\frac{1}{x}}}$ at $x=0$

- **Infinite Discontinuity :**

If at least one of the limits $\underset{x \to a^+}{Lt} f(x)$ and $\underset{x \to a^-}{Lt} f(x)$ be $\pm\infty$, then $f(x)$ has infinite discontinuity at $x=a$.

Example: $f(x) = \frac{\cos x}{x}$ at $x=0$.

- **Oscillatory Discontinuity:**

The Limit oscillates between two finite quantities

Example: $f(x) = \sin\frac{1}{x}$ at $x=0$.

➢ In case of discontinuity of the second kind, the absolute difference between the value of the RHL at $x = a$ and LHL at $x = a$ is called the Jump of Discontinuity. A function having a finite number of jumps in a given interval I is called a Piece wise continuous or Sectionally continuous function in this interval.

W.E-4:

The jump of discontinuity of the function $f(x) = \frac{|2x-3|}{2x-3}$ is

Solution :

$$f\left(\frac{3}{2}+\right) = 1, f\left(\frac{3}{2}-\right) = -1$$

$\therefore$ Jump of discontinuity $= 2$

➢ All polynomials, Trigonometrical functions, exponential and Logarithmic functions are continuous in their respective domains.

Intermediate value theorem :

➢ Suppose $f(x)$ is continuous on a interval I, and a , b are any two points of I. If y_0 is a number between $f(a)$ and $f(b)$, then there exists a number c between a and b such that $f(c) = y_0$.

Single Point Continuity:

➢ Functions which are continuous only at one point are said to exhibit single point continuity behaviour.

Example 1: $f(x) = \begin{cases} x, & \text{if } x \in Q \\ -x, & \text{if } x \notin Q \end{cases}$. is continuous only at $x = 0$.

Example 2: $f(x) = \begin{cases} x & , \text{if } x \in Q \\ 1-x & , \text{if } x \notin Q \end{cases}$ is continuous only at $x = 1/2$

W.E-5:

If $y = \frac{1}{t^2+t-2}$ where $t = \frac{1}{x-1}$, then the number of points of discontinuous of $y = f(x), x \in R$ is

Solution:

$t = \frac{1}{x-1}$ is discontinuous at x = 1. Also

$y = \frac{1}{t^2+t-2}$ is discountinuous at t = -2 and t = 1

When $t = -2$, $\frac{1}{x-1} = -2 \Rightarrow x = \frac{1}{2}$,

When $t = 1$, $\frac{1}{x-1} = 1 \Rightarrow x = 2$,

So, $y = f(x)$ is discontinuous at three points, $x = 1, \frac{1}{2}, 2$

C.U.Q

1. $|x|$ **is continuous at**

1) everywhere of x 2) $x = 0$ only

3) $(1, \infty)$ only 4) $(-\infty, 1)$ only

2. $Sgn(x)$ **is discontinuous at**

1) everywhere of x 2) $x = 0$

3) $(1, \infty)$ 4) $(-\infty, 1)$

3. $[x]$ **is continuous at**

1) $R - Z$ 2) Z 3) N 4) W

4. $\{x\}$ **is discontitinuous at (here $\{\ \}$ is a fractional part function)**

1) every where of x 2) Z

3) N 4) R-Z

C.U.Q - KEY

1) 1 2) 2 3) 1 4) 4

C.U.Q - HINTS

1) By using the graph

2) function $= \begin{cases} \frac{|x|}{x} & \text{at } x \neq 0 \\ 0 & \text{at } x = 0 \end{cases}$

3) By using the graph

4) By using the graph

LEVEL - I (C.W)

Continuity at a point :

1. The function

$$f(x)=\begin{cases}\dfrac{\cos 3x-\cos 4x}{x^2}, & for\ x\neq 0\\ \dfrac{7}{2} & , for\ x=0\end{cases}$$

at $x=0$ is (EAM - 2007)

1) continuous 2) discontinuous
3) left continuous 4) right continuous

2. The function defined by

$$f(x)=\begin{cases}x.\sin\dfrac{1}{x} & \text{for } x\neq 0\\ 0 & \text{for } x=0\end{cases}$$ at x = 0 is

1) continuous 2) right continuous
3) left continuous 4) can not be determined

3. If the function f(x) = $\dfrac{e^{x^2}-\cos x}{x^2}$ for $x\neq 0$ is continuous at $x=0$ then $f(0)=$

1) 1/2 2) 3/2 3) 2 4) 1/3

4. The value of f(0) so that the function

$$f(x)=\frac{\log\left(1+\frac{x}{a}\right)-\log\left(1-\frac{x}{b}\right)}{x},(x\neq 0)$$ is

continuous at $x=0$ is

1) $\dfrac{a+b}{ab}$ 2) $\dfrac{a-b}{ab}$ 3) $\dfrac{ab}{a+b}$ 4) $\dfrac{ab}{a-b}$

5. $f(x)=x\left[3-\log\left(\dfrac{\sin x}{x}\right)\right]-2$ to be continuous at $x=0$, then $f(0)=$

1) 0 2) 2 3) -2 4) 3

6. If $f(x)=x^{\frac{1}{x-1}}$ for $x\neq 1$ and f is continuous at $x=1$ then $f(1)=$

1) e 2) e^{-1} 3) e^{-2} 4) e^2

7. If $f(x)=\begin{cases}\dfrac{1-\sqrt{2}\sin x}{\pi-4x}, & if\ x\neq\dfrac{\pi}{4}\\ a & , if\ x=\dfrac{\pi}{4}\end{cases}$

is continuous at $x=\dfrac{\pi}{4}$ then $a=$

1) 4 2) 2 3) 1 4) 1/4

8. The discontinuous points of $f(x)=\dfrac{1}{\log|x|}$ are

1) $0,\pm 2$ 2) $1,\pm 2$ 3) $0,\pm 1$ 4) $0,\pm 3$

9. Let $f:R\to R$ be defined by

$$f(x)=\begin{cases}\alpha+\dfrac{\sin[x]}{x} & ,if\ x>0\\ 2 & ,if\ x=0\\ \beta+\left[\dfrac{\sin x-x}{x^3}\right] & ,if\ x<0\end{cases}$$ where

$[x]$ denotes the integral part of x. If f continuous at $x=0$, then $\beta-\alpha=$ [EAM - 2012]

1) -1 2) 1 3) 0 4) 2

10. If $[x]$ denotes a greatest integer not exceeding x and if the function f defined

by $f(x)=\begin{cases}\dfrac{a+2Cosx}{x^2}, & if\ x<0\\ b\tan\dfrac{a}{[x+4]}, & if\ x\geq 0\end{cases}$ is

continuous at x=0, then the ordered pair (a,b) is [EAM - 2011]

1) $(-2,1)$ 2) $(-2,-1)$
3) $(-1,\sqrt{3})$ 4) $(-2,-\sqrt{3})$

Continuity in an interval:

11. If $f:R\to R$ is defined by

$$f(x)=\begin{cases}\dfrac{x+2}{x^2+3x+2} & if\ x\in R-\{-1,-2\}\\ -1 & if\ x=-2\\ 0 & if\ x=-1\end{cases}$$

then f is continuous on the set [EAM-05]

1) R 2) $R-\{-2\}$
3) $R-\{-1\}$ 4) $R-\{-1,-2\}$

12. Let f be a non-zero real valued continuous function satisfying $f(x+y)=f(x).f(y)$ for all $x,y\in R$. If f(2)=9, then f(6)= (EAM-2013)

1) 3^2 2) 3^6 3) 3^4 4) 3^3

13. **If** $f:[-2,2]\to R$ **is defined by**

$$f(x)=\begin{cases}\dfrac{\sqrt{1+cx}-\sqrt{1-cx}}{x}, & for \quad -2\le x<0\\ \dfrac{x+3}{x+1}, & for \quad 0\le x\le 2\end{cases}$$

is continuous on $[-2,2]$ **then c =** **(EAM-2014)**

1) 3 2) $\frac{3}{2}$ 3) $\frac{3}{\sqrt{2}}$ 4) $\frac{2}{\sqrt{3}}$

14. $f(x)=\dfrac{p+q^{\frac{1}{x}}}{r+s^{\frac{1}{x}}}, s>1, q>1, r\neq 0$, $f(0)=1$ **is left continuous at** $x=0$ **then**

1) $p=0$ 2) $p=r$ 3) $p=q$ 4) $p\neq q$

15. **If** $f(x)=\begin{cases}\sin x, & if\ x \text{ is rational}\\ \cos x, & if\ x \text{ is irrational}\end{cases}$

then the function is

1) discontinuous at $x=n\pi+\frac{\pi}{4}$

2) continuous at $x=n\pi+\frac{\pi}{4}$

3) discontinuous at all x

4) continuous at all x

16. $f(x)=|x|+|x-1|$ **is continuous at**

1) 0 only 2) 0,1 only

3) every where 4) no where

LEVEL-I (C.W)-KEY

1) 1	**2) 1**	**3) 2**	**4) 1**
5) 3	**6) 1**	**7) 4**	**8) 3**
9) 2	10) 2	**11) 3**	**12)2**
13)1	**14) 2**	**15) 2**	**16) 3**

LEVEL-I (C.W)-HINTS

1. $\lim\limits_{x\to 0}\dfrac{\cos ax-\cos bx}{x^2}=\dfrac{b^2-a^2}{2}$ and $\lim\limits_{x\to 0}\dfrac{\sin x}{x}=1$

2. $f(x)$ is continuous at $x=0$

3. Applying L-Hospital rule

$$f(0)=\lim_{x\to 0}\frac{e^{x^2}.2x+\sin x}{2x}=\frac{3}{2}$$

4. Use L-Hospital rule

5. $f(0)=-2$

6. $f(1)=e^{\lim\limits_{x\to 1}\frac{1}{x-1}(x-1)}$

7. Use L-Hospital rule

8. $f(x)$ is discontinuous at $x=0,-1,1$

9. $\alpha+0=2=\beta-1$

$\alpha=\beta-1\Rightarrow\beta-\alpha=1$

10. $f(0)=b$

to define $f(0^-)$, $a=-2\Rightarrow b=-1$.

11. $f(x)$ is not continuous at -1

12. f(x)= 3^x

13. $f(0+)=f(0-)$

14. Put $x=-h$ as $h\to 0$

15. $\sin x=\cos x\Rightarrow x=n\pi+\frac{\pi}{4}$

16. Everywhere

LEVEL - I (H.W)

Continuity at a point :

1. **The function** $f(x)=\dfrac{x\tan 2x}{\sin 3x.\sin 5x}$, **for** $x\neq 0$, **is continuous at** $x=0$, **then** $f(0)$

1) $\frac{2}{13}$ 2) $\frac{2}{17}$ 3) $\frac{2}{11}$ 4) $\frac{2}{15}$

2. **If f (x)=**$\dfrac{1-\cos ax}{1-\cos bx}$, **for** $x\neq 0$, **is continuous at** $x=0$ **then** $f(0)=$

1) $\frac{a^2}{2}$ 2) $\frac{a}{b^2}$ 3) $\frac{a}{b}$ 4) $\left(\frac{a}{b}\right)^2$

3. **If the function** $f(x)=\begin{cases}\dfrac{\sin^2 ax}{x^2}, & for \quad x\neq 0\\ 1, & for \quad x=0\end{cases}$

is continuous at $x=0$ **then** $a=$

1) ± 1 2) 0 3) $\pm\frac{1}{2}$ 4) $\pm\frac{1}{3}$

4. **If the function** $f(x)=\begin{cases}\dfrac{2^{x+2}-16}{4^x-16}, & for\ x\neq 2\\ A, & for\ x=2\end{cases}$

is continuous at $x=2$, **then** $A=$

1) 2 2)1/2 3) 1/4 4) 0

5. If $f(x)=\begin{cases}\dfrac{(e^{kx}-1).\sin kx}{x^2}, & for\ x\neq 0\\ 4 & , for\ x=0\end{cases}$

is continuous at $x=0$ then k =

1) ± 1 2) ± 2 3) 0 4) ± 3

6. Let $f(x)=\dfrac{x+x^2+....+x^n-n}{x-1}$, $x\neq 1$, the value of $f(1)$ so that f is continuous at x=1 is

1) n 2) $\dfrac{n+1}{2}$

3) $\dfrac{n(n+1)}{2}$ 4) $\dfrac{n(n-1)}{2}$

7. The set of points of discontinuity of the function $f(x)=\dfrac{1}{x^2+x+1}$

1) ϕ 2) R 3) $\{0\}$ 4) R^-

8. If $f:R\to R$ defined by

$$f(x)=\begin{cases}\dfrac{1+3x^2-\cos 2x}{x^2}, & for\quad x\neq 0\\ k & , for\quad x=0\end{cases}$$

is continuous at $x=0$, then $k=$ [EAM-2010]

1) 1 2) 5 3) 6 4) 0

9. If $f:R\to R$ defined by

$$f(x)=\begin{cases}\dfrac{2\sin x-\sin 2x}{2x\cos x} & ,if\quad x\neq 0\\ a & ,if\quad x=0\end{cases}$$ then

the value of a so that f is continuous at x=0 is [EAM - 2009]

1) 2 2) 1 3) -1 4) 0

10. If $f(x)=\begin{cases}\dfrac{\sqrt{1+kx}-\sqrt{1-kx}}{x} & , for\ -1\leq x<0\\ 2x^2+3x-2 & , for\ 0\leq x\leq 1\end{cases}$

is continuous at $x=0$ then $k=$

1) –4 2) -3 3) -2 4) -1

Continuity in an interval:

11. The interval on which f(x)=$\sqrt{1-x^2}$ is continuous in

1) $(0,\infty)$ 2) $(1,\infty)$

3) [-1,1] 4) $(-\infty,-1)$

12. The function $f(x)=\begin{cases}0, & if\ x\ is\ irrational\\ 1, & if\ x\ is\ rational\end{cases}$ is

1) continuous at $x=1$
2) discontinuous only at 0
3) discontinuous only at 0,1
4) discontinuous everywhere

Continuity of special type of functions :

13. If $f(x)$ is a continuous function, then $\lim\limits_{x\to 0} f(x)\dfrac{|x|}{x}$ exist if

1) $f(x)$ is a polynomial
2) $f(x)=ax^2+bx+c$
3) $f(x)=ax^2+bx$
4) $f(x)=ax+b$

14. The funcion f(x)=$\dfrac{|x-3|}{x-3}$ at $x=3$, is

1) Left continuous 2)Right continuous
3) continuous 4) discontinuous

LEVEL-I (H.W)-KEY

1) 4	2) 4	3) 1	4) 2
5) 2	6) 3	7) 1	8) 2
9) 4	10) 3	11) 3	12) 4
13) 3	14) 4		

LEVEL-I (H.W)-HINTS

1. $f(0)=k=\lim\limits_{x\to 0}\dfrac{x\tan 2x}{\sin 3x\sin 5x}=\dfrac{2}{15}$

2. $f(0)=\dfrac{a^2}{b^2}$

3. $\lim\limits_{x\to 0}\dfrac{\sin^2 ax}{x^2}=1$

4. $A=\lim\limits_{x\to 2} f(x)$

$$= \lim_{x\to 2} \frac{2^x.2^2-16}{4^x-16}$$

$$= \lim_{x\to 2} \frac{2^2.2^x \log 2}{4^x.\log 4}$$

$$= \frac{4^2.\log 2}{4^2.\log 4} = \log_4 2 = \log_{2^2} 2^1 = \frac{1}{2}$$

5. $f(0) = \lim_{x\to 0}\left(\frac{e^{kx}-1}{x}\right).\frac{\sin kx}{x} \Rightarrow 4 = k^2$

6. $f(1) = \lim_{x\to 1} \frac{x+x^2+....+x^n-n}{x-1}\left(\frac{0}{0}\text{ from}\right)$

$$= \lim_{x\to 1} \frac{1+2x+....+nx^{n-1}}{1} = 1+2+...+n$$

$$= \frac{n(n+1)}{2}$$

7. $f(x)$ is defined for any $x \in R$

8. Use L-Hospital's Rule

9. Simplify

10. $f(x)$ is continuous at 0

11. $1-x^2 \geq 0;\ x \in [-1,1]$

12. If 'a' is any real number, then every neighbourhood of a contains infinitely many rationals as well as irrationals. So $f(x)$ assumes values 0 and 1 in every neighbourhood of a and hence $\lim_{x\to a} f(x)$ does not exist.
So, f is not continuous at any real number.

13. $\lim_{x\to 0^+} \frac{|x|}{x} = 1$ and $\lim_{x\to 0^-} \frac{|x|}{x} = -1$
Also $f(x)$ is continuous
$\therefore$ Given limit can exist only if
$\lim_{x\to 0} f(x) = 0$
$\therefore f(x) = ax^2 + bx$ is the only choice

14. $f(x)$ is not defined at $x = 3$

LEVEL - II (C.W)

Continuity at a point :

1. Let

$$f(x) = \begin{cases} \dfrac{(e^x-1)^{2n}}{\sin^n(x/a)(\log(1+(x/a)))^n}, & for\ x \neq 0 \\ 16^n, & for\ x = 0 \end{cases}$$

and f is a continuous at x=0, then the value of a is

1) 16 2) 2 3) 8 4) 4

2. If $f(x) = \begin{cases} x^\alpha \cos(1/x), & if\ x \neq 0 \\ 0, & if\ x = 0 \end{cases}$ **is continuous at** $x = 0$ **then**

1) $\alpha < 0$ 2) $\alpha > 0$ 3) $\alpha = 0$ 4) $\alpha \geq 0$

3. Let $f(x) = \begin{cases} \dfrac{x(1+a\cos x)-b\sin x}{x^3}, & if\ x \neq 0 \\ 1, & if\ x = 0 \end{cases}$

The values of a and b so that f is a continuous function at x=0, are

1) $5/2, 3/2$ 2) $5/2, -3/2$

3) $-5/2, -3/2$ 4) $-\frac{5}{2}, \frac{3}{2}$

4. If $f(x) = \begin{cases} a^2\cos^2 x + b^2 \sin^2 x, x \leq 0 \\ e^{ax+b}, x > 0 \end{cases}$

f(x) is continuous at $x = 0$ then

1) $2\log|a| = b$ 2) $2\log|b| = e$

3) $\log a = 2\log|b|$ 4) $a = b$

5. $f(x) = \begin{cases} \left(\dfrac{3}{x^2}\right)\sin 2x^2, & if\ x < 0 \\ \dfrac{x^2+2x+c}{1-3x^2}, & if\ x \geq 0, x \neq \dfrac{1}{\sqrt{3}} \\ 0, & if\ x = \dfrac{1}{\sqrt{3}} \end{cases}$

then in order that f be continuous at $x = 0$, the value of c is

1) 2 2) 4 3) 6 4) 8

Continuity in an interval:

6. If $f(x)=\begin{cases}-1, & if\ x<0\\ 0, & if\ x=0\\ 1, & if\ x>0\end{cases}$ and $g(x)=\sin x+\cos x$, then points of discontinuity of $(fog)(x)$ in $(0,2\pi)$ are

1) $\frac{\pi}{4},\frac{5\pi}{4}$ 2) $\frac{\pi}{4},\frac{3\pi}{4}$

3) $\frac{\pi}{4},\frac{7\pi}{4}$ 4) $\frac{3\pi}{4},\frac{7\pi}{4}$

Continuity of special type of functions :

7. If $f(x)$ be a continuous function for all real values of x and satisfies

$x^2+x(f(x)-2)+2\sqrt{3}-3-\sqrt{3}f(x)=0$

$\forall x\in R$ then the value of $f(\sqrt{3})$ is

1) $\sqrt{3}$ 2) $2(\sqrt{3}-1)$

3) $2\sqrt{3}-1$ 4) $2(1-\sqrt{3})$

8. Let f be a continuous function on R such that $f\left(\frac{1}{4n}\right)=\sin(e^n)e^{-n^2}+\frac{n^2}{n^2+1}$. Then the value of $f(0)$ is

1) 1 2) $\frac{1}{2}$ 3) 0 4) 2

9. The function $f(x)=a[x+1]+b[x-1]$, $(a\neq 0, b\neq 0)$ where $[x]$ is the greatest integer function is continuous at $x=1$ if

1) $a=2b$ 2) $a=b$

3) $a+b=0$ 4) $a+2b=0$

10. Let $f(x)=[2x^3-6]$ when $[x]$ is greatest integer less than or equal to x then the number of points in (1,2) where f is discontinuous is

1) 5 2) 7 3) 13 4) 12

11. If $f(x)=\begin{cases}\frac{\sin[x]}{[x]}, & for\ [x]\neq 0\\ 0 & ,for\ [x]=0\end{cases}$ is

1) continuous at $x=0$

2) discontinuous at $x=0$

3) L.H.L=0 4) R.H.L=1

12. If $x+2|y|=3y$, where $y=f(x)$, then $f(x)$ is

1) continuous everywhere

2) differentiable everywhere

3) discontinuous at $x=0$

4) Not differentiable at anywhere

13. $f(x)=\min\{x,x^2\}$, $\forall x\in R$ then $f(x)$ is

1) discontinuous at 0 2) discontinuous at 1

3) continuous on R 4) continuous at 0,1

LEVEL-II (C.W)-KEY

1) 4	2) 2	3) 3	4) 1
5) 3	6) 4	7) 4	8) 1
9) 3	10) 3	11) 2	12) 1
13) 3			

LEVEL-II (C.W)-HINTS

1. $\lim_{x\to 0}f(x)=$

$$\lim_{x\to 0}\left(\frac{e^x-1}{x}\right)^{2n}\frac{(x/a)^n}{\sin^n(x/a)}\times\frac{(x/a)^n}{(\log(1+(x/a)))^n}\cdot a^{2n}$$

$=a^{2n}$

since. $f(0)=\lim_{x\to 0}f(x)$ so

$a^{2n}=16^n=4^{2n}$ thus $a=4$.

2. Continuous when $\alpha>0$

3. Note that f is continuous everywhere except possibly at $x=0$. For f to be continuous at $x=0$,

$$1=f(0)=\lim_{x\to 0}\frac{x(1+a\cos x)-b\sin x}{x^3},\ \left(\frac{0}{0}\text{ form}\right)$$

$$=\lim_{x\to 0}\frac{(1+a\cos x)-xa\sin x-b\cos x}{3x^2}$$

$$=\lim_{x\to 0}\frac{1+(a-b)\cos x-xa\sin x}{3x^2}$$

This limit will exist only if $1+(a-b)=0$.

$$=\lim_{x\to 0}\frac{-(a-b)\sin x-ax\cos-a\sin x}{6x}$$

$$=\frac{1}{6}\lim_{x\to 0}\left[(b-2a)\frac{\sin x}{x}-a\cos x\right]$$

$$=\frac{1}{6}[b-3a]$$

$\Rightarrow b-3a=6$

Solving $a-b=-1$ and $b-3a=6$, we get $b=-3/2$ and $a=-5/2$.

4. $\lim_{x\to 0^-} f(x)=\lim_{x\to 0^+} f(x)$

5. $\lim_{x\to 0^-} f(x)=\lim_{x\to 0^-}\frac{3}{x^2}\sin 2x^2=6\lim_{x\to 0}\frac{\sin 2x^2}{2x^2}=6$

$\lim_{x\to 0^+} f(x)=\lim_{x\to 0^+}\frac{x^2+2x+c}{1-3x^2}=\frac{c}{1}=c$

Hence for f to be continuous $c=6$.

6. $$f(g(x))=\begin{cases}1, \text{ if } & 0<x<\frac{3\pi}{4} or \frac{7\pi}{4}<x<2\pi\\ 0, \text{ if } & x=\frac{3\pi}{4},\frac{7\pi}{4}\\ -1, \text{ if } & \frac{3\pi}{4}<x<\frac{7\pi}{4}\end{cases}$$

Clearly $(fog)(x)$ is not continuous at $x=\frac{3\pi}{4},\frac{7\pi}{4}$

7. $f(x)$ is continuous at $x=\sqrt{3}$

$\Rightarrow f(\sqrt{3})=\underset{x\to\sqrt{3}}{Lt}\frac{x^2-2x+2\sqrt{3}-3}{\sqrt{3}-x}$

8. $f(0)=\underset{x\to 0}{Lt} f(x)$

$=\underset{n\to\infty}{Lt} f\left(\frac{1}{4n}\right)=0+1=1$

9. $f(1)=2a,\ \lim_{x\to 1^-} f(x)=a-b$ so $a+b=0$ for f to be continuous at $x=1$.

10. $1<x<2 \ \Rightarrow 1<x^3<8$

$\Rightarrow 2<2x^3<16$

$\Rightarrow -4<2x^3-6<10$

$\therefore\ [2x^3-6]$ is discontinuous at -3,-2,-1,0,1,2,3,4,5,6,7,8,9

11. $L.H.L=\lim_{x\to 0^-}\frac{\sin(-1)}{-1}=\sin 1$

$(\because x\to 0^- \Rightarrow [x]=-1)$

$R.H.L=0\ \ (\because x\to 0^+\Rightarrow [x]=0)$

12. $x+2|y|=3y$

$\Rightarrow x+2y=3y, y\ge 0$ and

$x-2y=3y, y<0$

$$\Rightarrow y=\begin{cases}x, & x\ge 0\\ \frac{x}{5}, & x<0\end{cases}$$

Therefore, y is continuous everywhere but is not differentiable at $x=0$.

13. $x=x^2 \ \Rightarrow\ x=0,x=1$

$$f(x)=\begin{cases}x \text{ for } x<0\\ x^2 \text{ for } 0\le x\le 1\\ x \text{ for } x>1\end{cases}$$

LEVEL - II (H.W)

Continuity at a point :

1. If $f(x)=\frac{\sqrt{a^2-ax+x^2}-\sqrt{a^2+ax+x^2}}{\sqrt{a+x}-\sqrt{a-x}}$ is continuous at $x=0$ then $f(0)=$

1) $\sqrt{a}$ 2) $-\sqrt{a}$ 3) $a\sqrt{a}$ 4) $-a\sqrt{a}$

2. If $f(x)=\begin{cases}\frac{72^x-9^x-8^x+1}{\sqrt{2}-\sqrt{1+\cos x}}, & if\ x\ne 0\\ K\log 2\log 3, & if\ x=0\end{cases}$ is a continuous function, then K is equal to

1) $\sqrt{2}$ 2) 24 3) $18\sqrt{3}$ 4) $24\sqrt{2}$

Continuity in an interval:

3. The value of f(0) so that the function f(x)=$\frac{1-\cos(1-\cos x)}{x^4}$ is continuous everywhere is

1) $\frac{1}{8}$ 2) $\frac{1}{2}$ 3) $\frac{1}{4}$ 4) $\frac{1}{3}$

4. **The function $y = 3\sqrt{x} - |x-1|$ is continuous at**

1) $x<0$ 2) $x \geq 1$
3) $0 \leq x \leq 1$ 4) $x \geq 0$

5. **If the function** $f(x)=\begin{cases} \frac{1-\cos 4x}{x^2}, & if\ x<0 \\ a, & if\ x=0 \\ \frac{\sqrt{x}}{\sqrt{16+\sqrt{x}}-4}, & if\ x>0 \end{cases}$ **is continuous at $x=0$ then $a=$**

1) 8 2) $\frac{1}{8}$ 3) -8 4) 0

6. $f(x)=\begin{cases} \frac{(x+bx^2)^{1/2}-x^{1/2}}{bx^{3/2}}, & if\ x>0 \\ c, & if\ x=0 \\ \frac{Sin(a+1)x+\sin x}{x}, & if\ x<0 \end{cases}$

is continuous at $x=0$ then

1) $a=\frac{-3}{2}, b=0, c=\frac{1}{2}$

2) $a=\frac{-3}{2}, b\neq 0, c=\frac{1}{2}$

3) $a=\frac{3}{2}, b\neq 0, c=\frac{1}{2}$

4) $a=\frac{3}{2}, b\neq 0, c=-\frac{1}{2}$

7. **If** $f(x)=\begin{cases} (1+|\sin x|)^{\frac{a}{|\sin x|}}, & if\ -\frac{\pi}{6}<x<0 \\ b, & if\ x=0 \\ e^{\frac{\tan 2x}{\tan 3x}}, & if\ 0<x<\frac{\pi}{6} \end{cases}$ **is continuous at x = 0 then**

1) $a=e^{2/3}, b=2/3$ 2) $a=2/3, b=e^{2/3}$

3) $a=1/3, b=e^{1/3}$ 4) $a=e^{1/3}, b=e^{1/3}$

Continuity of special type of functions :

8. **Let $f(x)=g(x)\frac{e^{1/x}-e^{-1/x}}{e^{1/x}+e^{-1/x}}$, where g is a continuous function then $\lim_{x\to 0} f(x)$ exist if**

1) $g(x)=x+2$ 2) $g(x)=x^2+4$

3) $g(x)=xh(x), h(x)$ is a polynomial

4) $g(x)$ is a constant function

9. The function f(x)=$[\cos x]$ is

1) continuous at $x=\frac{\pi}{2}$

2) discontinuous at $x=\frac{\pi}{2}$

3) L.H.L= -1 at $x=\frac{\pi}{2}$

4) R.H.L= 1 at $x=\frac{\pi}{2}$

10. **The function $f(x)=\frac{1}{x^2-3|x|+2}$ is discontinuous at the points**

1) $x=1, 2$ 2) $x=\pm 1, \pm 2$
3) R 4) $R-\{1,2\}$

11 **Let a function f be defined by $f(x)=\frac{x-|x|}{x}$ for $x\neq 0$ and $f(0)=2$, then f is**

1) continuous no where
2) continuous every where
3) continuous for all x except $x=1$
4) continuous for all x except $x=0$

12. **If $f(x)=\frac{1}{1-x}$ then the points of discontinuity of $(f0f0f)(x)$ is**

1) $\{0,1\}$ 2) $\{0,\pm 1\}$ 3) $\{1\}$ 4) $\{\pm 1\}$

LEVEL-II (H.W)-KEY

1) 2	2) 4	3) 1	4) 4
5) 1	6) 2	7) 2	8) 3
9) 2	10) 2	11) 4	12) 1

LEVEL-II (H.W)-HINTS

1. $f(0)=\lim_{x\to 0} f(x)$ and rationalise or use L.H. rule

2. $K\log 2\log 3 = f(0)$

$$=\lim_{x\to 0} f(x)=\lim_{x\to 0}\frac{72^x-9^x-8^x+1}{\sqrt{2}-\sqrt{1+\cos x}}$$

$$=\lim_{x\to 0}\frac{(9^x-1)(8^x-1)}{x^2}\frac{x^2}{\sqrt{2}(1-\cos x/2)}$$

$$=\lim_{x\to 0}\frac{9^x-1}{x}\cdot\frac{8^x-1}{x}\frac{16(x/4)^2}{\sqrt{2}\cdot 2\sin^2 x/4}$$

$$=\frac{16}{2\sqrt{2}}\log 9\log 8$$

$$=\frac{8}{\sqrt{2}}6\log 3\log 2 = 24\sqrt{2}\log 3\log 2$$

Thus $K=24\sqrt{2}$.

3. $$\lim_{x\to 0}\frac{1-\cos(1-\cos x)}{(1-\cos x)^2}\frac{(1-\cos x)^2}{x^4}$$

$$=\lim_{x\to 0}\frac{1-\cos(1-\cos x)}{(1-\cos x)^2}\left(\frac{(1-\cos x)}{x^2}\right)^2$$

$$=\frac{1}{2}\times\frac{1}{4}\left(\lim_{x\to 0}\frac{1-\cos x}{x^2}=\frac{1}{2}\right)(formula)$$

$$=\frac{1}{8}$$

4. $$\lim_{x\to 0^+} f(x)=\lim_{x\to 0^+} 3\sqrt{x}-|x-1|$$

$=0-|-1|=-1$

Since $\lim_{x\to 0^+} f(x)=f(0)$

$\therefore f(x)$ is continuous for $x\ge 0$

5. $a=\lim_{x\to 0^-}\frac{1-\cos 4x}{x^2}=8$

6. $\lim_{x\to 0^-} f(x)=\lim_{x\to 0^-} f(x)=f(0)$

7. $\lim_{x\to 0^-} f(x)=\lim_{x\to 0^+} f(x)=f(0)$

8. $$\lim_{x\to 0^+}\frac{e^{1/x}-e^{-1/x}}{e^{1/x}+e^{-1/x}}=\lim_{x\to 0^+}\frac{1-e^{-2/x}}{1+e^{-2/x}}=1$$

$$\lim_{x\to 0^-}\frac{e^{1/x}-e^{-1/x}}{e^{1/x}+e^{-1/x}}=\lim_{x\to 0^-}\frac{e^{2/x}-1}{e^{2/x}+1}=-1$$

Hence $\lim_{x\to 0} f(x)$ exists if $g(x)=xh(x)$

where $h(x)$ is a continuous function. If

$g(x)=a(a\ne 0)$ then

$\lim_{x\to 0^+} f(x)=a,\ \lim_{x\to 0^-} f(x)=-a$. Thus

$\lim_{x\to 0} f(x)$ does not exist if $f(x)=x+2$

or x^2+4 or is a constant function.

9. $L.H.L=\lim_{x\to\frac{\pi}{2}^-}[\cos x]=0$

$R.H.L=\lim_{x\to\frac{\pi}{2}^+}[\cos x]=-1$

10. $f(x)$ is discontinuous when

$x^2-3|x|+2=0$

$\Rightarrow |x|^2-3|x|+2=0$

$\Rightarrow |x|=1,2$

11. $$f(x)=\begin{cases}2, & if\ x<0\\ 0, & if\ x>0\\ 2, & if\ x=0\end{cases}$$

Thus $\lim_{x\to 0^-} f(x)=2\ne 0=\lim_{x\to 0^+} f(x)$.

Hence, f is continuous every where except at $x=0$.

12. $f(x)=\frac{1}{1-x}$

$$(fof)(x)=\frac{1}{1-\frac{1}{1-x}}=\frac{1-x}{-x}=\frac{x-1}{x}$$

$$(fofof)(x)=\frac{1}{1-\frac{x-1}{x}}=x$$

$(fofof)(x)$ is discontinuous at $x=0, x=1$.

LEVEL - III

Continuity at a point :

1. If $f(x)=\begin{cases}\frac{|x+2|}{\tan^{-1}(x+2)}, & if\ x\neq -2\\ 2, & if\ x=-2\end{cases}$ then, $f(x)$ is
   1) continuous at $x=-2$
   2) not continuous at $x=-2$
   3) differentiable at $x=-2$
   4) continuous but not diferentiable at $x=-2$

2. Let $f(x)=\frac{(256+ax)^{1/8}-2}{(32+bx)^{1/5}-2}$. If f is continuous at $x=0$, then the value of a/b is
   1) $\frac{8}{5}f(0)$ 2) $\frac{32}{5}f(0)$
   3) $\frac{64}{5}f(0)$ 4) $\frac{16}{5}f(0)$

3. If $f''(x)$ be continuous at $x=0, f(0)\neq 0, f^{1}(0)\neq 0$ and $\lim_{x\to 0}\frac{2f(x)-3af(2x)+bf(8x)}{\sin^2 x}$, exists then the values of a and b are
   1) $a=\frac{7}{9}, b=\frac{1}{3}$ 2) $a=1, b=1$
   3) $a=b=-1$ 4) $a=-1, b=1$

4. The values of a and b if f is continuous at $x=0$, where
$$f(x)=\begin{cases}\left(1+\frac{ax+bx^3}{x^2}\right)^{1/x}, & if\ x>0\\ 3, & if\ x=0\end{cases}$$
   1) $a=0, b=\log 3$ 2) $a=1, b=\log 2$
   3) $a=2, b=\log 3$ 4) $a=0, b=\log 2$

5. Let $f(x)=\frac{\log(1+x+x^2)+\log(1-x+x^2)}{\sec x-\cos x}$, $x\neq 0$. Then the value of $f(0)$ so that f is continuous at $x=0$ is
   1) 1 2) 0 3) 2 4) -1

6. $If\ f(x)=\begin{cases}\frac{\sin(\cos x)-\cos x}{(\pi-2x)^3}, & if\ x\neq\frac{\pi}{2}\\ k, & if\ x=\frac{\pi}{2}\end{cases}$ is continuous at $x=\frac{\pi}{2}$, then k =
   1) 0 2) $-\frac{1}{6}$ 3) $-\frac{1}{24}$ 4) $-\frac{1}{48}$

7. If $f(x)=\lim_{n\to\infty}\frac{\log(2+x)-x^{2n}\sin x}{1+x^{2n}}$ then $f(x)$ is discontinuous at
   1) $x=1$ only 2) $x=-1$ only
   3) $x=-1,1$ only 4) no point

8. Let $f:R\to R$ be given by
$f(x)=\begin{cases}5x, & \text{if } x\in Q\\ x^2+6, & \text{if } x\in R-Q\end{cases}$ then
   1) f is continuous at $x=2$ and $x=3$
   2) f is discontinuous at $x=2$ and $x=3$
   3) f is contininuous at $x=2$ but not at $x=3$
   4) f is continuous at $x=3$ but not at $x=2$

Continuity in an interval:

9. $If\ f(x)=\begin{cases}\frac{A+3\cos x}{x^2}, & if\ x<0\\ B\tan\left(\frac{\pi}{[x+3]}\right), & if\ x\geq 0\end{cases}$ Where $[.]$ represents the greatest integer function, is continuous at $x=0$ Then
   1) $A=-3, B=-\sqrt{3}$ 2) $A=3, B=-\frac{\sqrt{3}}{2}$
   3) $A=-3, B=-\frac{\sqrt{3}}{2}$ 4) $A=-\frac{\sqrt{3}}{2}, B=-3$

10. If $f(x)=\begin{cases}\frac{1-\sin^3 x}{3\cos^2 x}, & if\ x<\frac{\pi}{2}\\ a, & if\ x=\frac{\pi}{2},\\ \frac{b(1-\sin x)}{(\pi-2x)^2}, & if\ x>\frac{\pi}{2}\end{cases}$ so that $f(x)$ is continuous at $x=\frac{\pi}{2}$, then
   1) $a=1/2$ 2) $b=5$
   3) $a=1$ 4) $b=-4$

11. If $f(x)=\begin{cases}\dfrac{a|x^2-15x+56|}{x-8} & ,if\ x>9\\ b & ,if\ x=9\\ \dfrac{x-[x]}{x-8} & ,if\ x<9\end{cases}$,

where [.] denotes greatest integer function and the function is continuous then

1) $a=\frac{1}{2},b=1$ 2) $a=0,b=1$

3) $a=\frac{-1}{2},b=1$ 4) $a=\frac{-1}{2},b=-1$

12. If $f(x)=\begin{cases}bx^2-a & ,\ if\ x<-1\\ ax^2-bx-2, & if\ x\geq -1\end{cases}$, **If $f^1(x)$ is continuous everywhere. Then the equation whose roots are a and b is**

1) $x^2+3x-2=0$ 2) $x^2-3x+2=0$

3) $x^2+3x+2=0$ 4) $x^2-5x+6=0$

13. **Let** $f(x)=\dfrac{\log(1+x^2)}{x^4-26x^2+25}$. **Then**

1) f is continuous on $[6,10]$
2) f is continuous on $[-2,2]$
3) f is continous on $[-6,6]$
4) f is continuous on $[1,7]$

14. **If $[.]$ denotes the greatest integer function then the number of points where**

$f(x)=[x]+\left[x+\frac{1}{3}\right]+\left[x+\frac{2}{3}\right]$ **is discontinuous for $x\in(0,3)$ are**

1) 2 2) 9 3) 8 4) 10

15. $f(x)=\begin{cases}x[x] & ,if\ o\leq x<3\\ (x-1)[x], & if\ 3\leq x\leq 4\end{cases}$ **where $[x]$ is the greatest integer function. The function $f(x)$ is**

1) Continuous and differentiable at $x=3$
2) Continuous at $x=1,2,3$
3) Continuous but not differentiable at $x=3$
4) discontinuous at $x=1,2,3$

16. **If the function**

$$f(x)=\left[\frac{(x-5)^3}{A}\right]\sin(x-5)+a\cos(x-2)$$

where $[.]$ denotes the greatest integer function, is continuous and differentiable in (7,9), then

1) $A\in[8,64]$ 2) $A\in(0,8]$

3) $A\in[64,\infty)$ 4) $A\in[8,16]$

17. **The function**

$f(x)=\begin{cases}x^2-ax+3, & if\ x\text{ is rational}\\ 2-x, & \text{if } x \text{ is irrational}\end{cases}$, **is continuous at exactly two points then the possible values of '*a*' are**

1) $(2,\infty)$ 2) $(-\infty,3)$

3) $(-\infty,-1)\cup(3,\infty)$ 4) R

18. **A point where the function $f(x)$ is not continuous when $f(x)=[\sin[x]]$ in $(0,2\pi)$ ([.] denotes greatest integer $\leq x$) is**

1) $(3,0)$ 2) $(2,0)$ 3) $(1,0)$ 4) $(4,-1)$

19. **Let $f(x)=\text{sgn}(x)$ and $g(x)=x(x^2-5x+6)$. The function $f(g(x))$ is discontinuous at**

1) infinitely many points 2) exactly one point
3) exactly three points 4) no point

20. **Consider the function $f(x)=x-|x-x^2|$, $-1\leq x\leq 2$. The points of discontinuities of $f(x)$ for $x\in[-1,2]$ are**

1) $x=0,1$ 2) $x=1,2$

3) $x=0,\frac{1}{2},1$ 4) no points of discontinuities

21. If the function f(x)=$\dfrac{\sqrt{x^2+kx+1}}{x^2-k}$ is Continuous for every $x\in R$ Then

1) $k\in[-2,0)$ 2) $k\in(0,\infty)$

3) $k\in(-\infty,0)$ 4) $k\in R$

Continuity of special type of functions :

22. **The function defined by** $f(x)=(-1)^{[x^3]}$ **where** $[.]$ **denotes the greatest integer function, is**

1) Discontinuous for $x=n^{\frac{1}{3}}$ where n is any integer

2) $f\left(\frac{3}{2}\right)=1$

3) $f^1(x)=1 \text{ for } -1<x<1$ 4) Continuous on R

23. **If** $f(x)=\begin{cases} xe^{-\left(\frac{1}{|x|}+\frac{1}{x}\right)}, & if\ x\neq 0 \\ 0, & if\ x=0 \end{cases}$**, then** $f(x)$ **is**

1) continuous for all x, but is not differentiable
2) neither differentiable nor continuous
3) discontinuous everywhere
4) continuous as well as differentiable for all x

24. **The function** $f(x)=\cos^{-1}(\cos x)$ **is**

1) Continuous at $x=\pi$
2) Discontinuous $x=\pi$
3) Discontinuous at $x=-\pi$
4) $f(x)=x, \forall x \in R$

25. **The function** $f(x)=[x]^2-[x^2]$ **(where [y] is the largest integer** $\leq y$**) is discontinuous at**

1) all integers
2) all integers except 0 and 1
3) all integers except 0
4) all integers except 1

26. **If** $f(x)=\frac{1}{x^2-17x+66}$ **then** $f\left(\frac{2}{x-2}\right)$ **is discontinuous at x is equal to**

1) $2,\frac{7}{3},\frac{25}{11}$ 2) $2,\frac{8}{3},\frac{24}{11}$

3) $2,\frac{7}{3},\frac{24}{11}$ 4) $2,6,11$

27. **The function** $f(x)=|2\,\text{sgn}(2x)|+2$ **has**

1) Jump discontinuity
2) Removable discontinuity
3) Infinite discontinuity
4) No discontinuity at $x=0$

28. $f(x)=Sgn(x^3-x)$ **is discontinuous at** $x=$

1) 0 2) 1 3) - 1 4) 0, -1, 1

29. **If** $f(x)=Sgn(2\sin x+a)$ **is continuous for all** x **then the possible values of '** a **' are**

1) R 2) $a<-2$ or $a>2$
3) $(-2,2)$ 4) $(0,\infty)$

30. **If** $f:R\to R$ **is a function defined by** $f(x)=[x]\cos\left(\frac{2x-1}{2}\right)\pi$, **where [x] denotes the greatest integer function, then** f **is** **[AIEEE - 2012]**

1) Continuous for every real x
2) discontinuous only at $x=0$
3) discontinuous only at non-zero integral values of x.
4) continuous only at $x=0$.

31. **The values of** p **and** q **for which the function**

$$f(x)=\begin{cases} \frac{\sin(p+1)x+\sin x}{x}, & if\ x<0 \\ q, & if\ x=0 \\ \frac{\sqrt{x+x^2}-\sqrt{x}}{x^{\frac{3}{2}}}, & if\ x>0 \end{cases}$$ **is**

continuous for all x **in** R**, is [AIEEE - 2011]**

1) $p=\frac{5}{2},q=\frac{1}{2}$ 2) $p=-\frac{3}{2},q=\frac{1}{2}$

3) $p=\frac{1}{2},q=\frac{3}{2}$ 4) $p=\frac{1}{2},q=-\frac{3}{2}$

32. If $f(x)=\underset{x\to\infty}{Lt}(\sin x)^{2n}$, then f(x) is

1) discontinuous at $x=\pi/2$
2) continous at $x=\pi/2$
3) discontinuous at $x=-\pi/2$
4) discontinuous at infinite number of points

LEVEL-III-KEY

1) 2	2) 3	3) 1	4) 1	5) 1	6) 4
7) 3	8) 1	9) 3	10) 1	11) 1	12) 2
13) 1	14) 3	15) 3	16) 3	17) 3	18) 4
19) 3	20) 4	21) 1	22) 1	23) 1	24) 1
25) 4	26) 3	27) 2	28) 4	29) 2	30) 1
31) 2	32) 4				

LEVEL-III-HINTS

1. $\lim_{x\to -2^-} f(x) = \lim_{x\to -2^-} \frac{|x+2|}{\tan^{-1}(x+2)}$

$= \lim_{x\to -2^-} \frac{-(x+2)}{\tan^{-1}(x+2)} = -1$

$\left[\because \lim_{x\to 0^-} \frac{\tan^{-1} x}{x} = 1\right]$

$\lim_{x\to -2^+} f(x) = \lim_{x\to -2^+} \frac{|x+2|}{\tan^{-1}(x+2)}$

$= \lim_{x\to -2^+} \frac{x+2}{\tan^{-1}(x+2)} = 1$

$\therefore \lim_{x\to -2} f(x)$ does not exist.

2. $\lim_{x\to 0} \frac{(256+ax)^{1/8}-2}{(32+bx)^{1/5}-2} = f(0)$

Use L-Hospital rule

$\Rightarrow \lim_{x\to 0} \frac{\frac{1}{8}(256+ax)^{\frac{-7}{8}}.a}{\frac{1}{5}(32+bx)^{\frac{-4}{5}}.b} = f(0)$

On simplify, $\frac{a}{b} = \frac{64}{5}.f(0)$

3. Use L-Hospital rule and Equating $Nr = 0$

then, $3a-b-2=0 \to (1)$

$\Rightarrow 3a-4b-1=0 \to (2)$

Solving (1) and (2), $a=\frac{7}{9}, b=\frac{1}{3}$

Solving (1) and (2), $a=\frac{7}{9}, b=\frac{1}{3}$

4. $\lim_{x\to 0^+} f(x) = f(0)$

$\Rightarrow \lim_{x\to 0^+} \left(1+\frac{ax+bx^3}{x^2}\right)^{1/x} = 3$

$\Rightarrow \lim_{x\to 0} \left(1+\frac{a}{x}+bx\right)^{1/x} = 3$

L.H.S. is exist when $a=0$

$\Rightarrow \lim_{x\to 0}(1+bx)^{1/x} = 3$

$\Rightarrow e^6 = 3 \Rightarrow b = \log_e^3$

5. For f to be continuous at $x=0$

$f(0) = \lim_{x\to 0} f(x)$

$= \lim_{x\to 0} \frac{\log(1+x+x^2)+\log(1-x+x^2)}{\sec x - \cos x}$

$= \lim_{x\to 0} \frac{\log\left((1+x^2)^2 - x^2\right)}{1-\cos^2 x}\cos x$

$= \lim_{x\to 0} \frac{\log\left(1+(x^2+x^4)\right)}{\sin^2 x}\cos x$

$= \lim_{x\to 0} \frac{\log\left(1+(x^2+x^4)\right)}{x^2+x^4} \times \frac{x^2}{\sin^2 x}$

$\times (1+x^2)\cos x = 1$

6. $k = \lim_{x\to \frac{\pi}{2}} \frac{\sin(\cos x)-\cos x}{(\pi-2x)^3}$

$\Rightarrow k = \lim_{x\to\frac{\pi}{2}} \frac{\sin(\cos x)-\cos x}{(\cos x)^3} \times \frac{\sin^3\left(\frac{\pi}{2}-x\right)}{8\left(\frac{\pi}{2}-x\right)^3}$

$\Rightarrow k = -\frac{1}{6} \times \frac{1}{8} = -\frac{1}{48}$

$\left[\therefore \lim_{x\to 0} \frac{\sin x - x}{x^3} = -\frac{1}{6}\right]$

7. $f(x)$ is discontinuous at $x=1$ or $x=-1$

8. f is continuous when $5x = x^2+6$

$\Rightarrow x^2-5x+6=0$

$\Rightarrow x = 2,3$

9. $\lim_{x\to 0^-} f(x) = \lim_{x\to 0} \left(\frac{A+3}{x^2} - \frac{3}{2!} + \frac{3}{4!}x^2 - \frac{3}{6!}x^4 + ...\right)$

For this limit to exist, we must have $A+3=0$ and in that case, we have

$\lim_{x\to 0^-} f(x) = -\frac{3}{2}$ Now,

$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} B\tan\left(\frac{\pi}{[x+3]}\right) = B\tan\frac{\pi}{3} = B\sqrt{3}$

$\lim_{x\to 0^-} f(x) = \lim_{x\to 0^+} f(x) = f(0)$

10. $\Rightarrow \lim_{x\to\frac{\pi}{2}^-} \frac{1-\sin^3 x}{3\cos^2 x} = \lim_{x\to\frac{\pi}{2}^+} \frac{b(1-\sin x)}{(\pi-2x)^2} = a$

By L-Hospital rule

$\Rightarrow \lim_{x\to\frac{\pi}{2}^-} \frac{+3\sin^2 x.\cos x}{+6\cos x\sin x} = \lim_{x\to\frac{\pi}{2}^+} \frac{-b\cos x}{2(\pi-2x)(-2)} = a$

$\Rightarrow \lim_{x\to\frac{\pi}{2}^-} \frac{\sin x}{2} = \lim_{x\to\frac{\pi}{2}^+} \frac{b\sin x}{8} = a$

$\Rightarrow \frac{1}{2} = \frac{b}{8} = a \quad \therefore a = \frac{1}{2}$

11. $\lim_{x\to 9^+} f(x) = \lim_{x\to 9^-} f(x) = f(9)$

$\Rightarrow \lim_{x\to 9^+} \frac{a|x^2-15x+56|}{x-8} = \lim_{x\to 9^-} \frac{x-[x]}{x-8} = b$

$\Rightarrow \lim_{x\to 9^+} \frac{a(x-7)(x-8)}{x-8} = \lim_{x\to 9^-} \frac{x-8}{x-8} = b$

$\Rightarrow 2a = 1 = b$

12. $\lim_{x\to -1^-} f'(x) = \lim_{x\to -1^+} f'(x)$

$\Rightarrow \lim_{x\to -1^-} 2bx = \lim_{x\to -1^+} 2ax - b$

$\Rightarrow -2b = -2a - b$

$\Rightarrow -b = -2a \Rightarrow 2a = b$

$\Rightarrow f(x)$ is also continuous

$\Rightarrow \lim_{x\to -1^-} bx^2 - a = \lim_{x\to -1^+} ax^2 - bx - 2$

$\Rightarrow b - a = a + b - 2 \Rightarrow 2a = 2$

$\Rightarrow a = 1 \Rightarrow b = 2$

Equation is $x^2 - 3x + 2 = 0$

13. $f(x) = \frac{\log(1+x^2)}{(x^2-25)(x^2-1)}$

$= \frac{\log(1+x^2)}{(x-5)(x+5)(x-1)(x+1)}$

since $\log(1+x^2)$ is continuous on $(-\infty,\infty)$, so f is continuous on $R \sim \{-5,-1,1,5\}$ which clearly contains the interval $[6, 10]$.

14. $f(x) = [3x]$ is discontinuous when $3x =$ an integer

$\Rightarrow x = \frac{1}{3},\frac{2}{3},\frac{3}{3},\frac{4}{3},\frac{5}{3},\frac{6}{3},\frac{7}{3},\frac{8}{3} \in (0,3)$

15. $f(x) = \begin{cases} 0 & ; \quad 0 \le x < 1 \\ x & ; \quad 1 \le x < 2 \\ 2x & ; \quad 2 \le x < 3 \\ 3(x-1) & ; \quad 3 \le x < 4 \\ 12 & ; \quad x = 4 \end{cases}$

16. $[x]$ is not continuous & differentiable at integral values (points) so $f(x)$ continuous and differentiable in (7,9) if $\left[\frac{(x-5)^3}{A}\right] = 0$

$\Rightarrow A \ge (9-5)^3$

$\Rightarrow A \ge 64 \quad \therefore A \in [64,\infty)$

17. $f(x)$ is continuous when $x^2 - ax + 3 = 2 - x$

$\Rightarrow x^2 - (a-1)x + 1 = 0$. This must have two distinct roots $\Rightarrow \Delta > 0$

$\Rightarrow (a-1)^2 - 4 > 0$

18. $x \in [1,4)$. The value of $f(x)$ is zero

$x \in [4,2\pi)$. The value of $f(x)$ is -1.

19. It is discontinuous at $x = 0,2,3$

20. It is everywhere continuous.

21. $x^2 + kx + 1 \ge 0$ and $x^2 - k$ must not have any real root

$\therefore k^2 - 4 \le 0$ & $k < 0$

$\Rightarrow k \in [-2,2]$ *and* $k < 0$

$\Rightarrow k \in [-2,0)$

22. $f(x)$ is discontinuous when $x^3 = n; n \in Z$

$\Rightarrow x = n^{\frac{1}{3}}$

$f\left(\frac{3}{2}\right) = (-1)^3 = -1$

$\left.\begin{array}{l} \text{for } x \in (-1,0)\ f(x) = (-1)^{-1} = -1 \\ \text{for } x \in [0,1)\ f(x) = (-1)^0 = 1 \end{array}\right\} \Rightarrow f^1(x) = 0$

23. $|x|$ is not differentiable at $x = 0$

$|x|$ is continuous at $x = 0$

24. $f(x)=\cos^{-1}(\cos x)=\begin{cases}2\pi+x & \text{for } -2\pi<x<-\pi\\ -x & \text{for } -\pi\le x<0\\ x & \text{for } 0\le x\le\pi\\ 2\pi-x & \text{for } \pi<x<2\pi\end{cases}$

$f(x)$ is continuous at $x=\pi,-\pi$

25. Clearly, $f(x)=0$ for each integral value of x.

Also, if $0<x<1$, then $0<x^2<1$,

$\Rightarrow [x]=0$ and $[x^2]=0$

$\therefore f(x)=0$ for $0<x<1$

Again, if $1\le x<\sqrt{2}$ then $1\le x^2<2$

$\Rightarrow [x]=1,[x^2]=1$

$\therefore f(x)=0$ for $1\le x<\sqrt{2}$

$\therefore f(x)=0$ is continuous at $x=1$

However, at points x other than integers and not lying between 0 and $\sqrt{2}$, $f(x)\ne 0$

26. $u=\dfrac{2}{x-2}$ is discontinuous at $x=2$

$f(u)=\dfrac{1}{u^2-17u+66}=\dfrac{1}{(u-11)(u-6)}$ is discontnuous at $u=6,11$

$\therefore \dfrac{2}{x-2}=6,11\Rightarrow x=\dfrac{7}{3},\dfrac{24}{11}$

27. $f(x)=|2\,\text{sgn}(2x)|+2$

$=\left|2.\dfrac{2x}{|2x|}\right|+2=\left|2.\dfrac{x}{|x|}\right|+2$

$f(0)=2$

$f(0^+)=|2\times 1|+2=4$

$f(0^-)=|2(-1)|+2=4$

$f(0^+)=f(0^-)\ne f(0)$

$\therefore f(x)$ has removable discontinuity

28. $f(x)$ is discontinuous when $x^3-x=0$

$\Rightarrow x=0,-1,1$

29. Function continuous for all x

$\Rightarrow 2\sin x+a\ne 0\Rightarrow \sin x\ne\dfrac{-a}{2}$

$\Rightarrow \left|\dfrac{a}{2}\right|>1\Rightarrow a<-2\,(\text{or})\,a>2$

30. $=[x]\sin\pi x$ is continuous for every real x.

31. Use L-Hospital's rule.

32. If

$f(x)=\underset{x\to\infty}{Lt}\left(\sin^2 x\right)^n$

$=\begin{cases}1,\ if\ x=(2n+1)\dfrac{\pi}{2},\forall n\in z\\ 0,\ if\ x\ne(2n+1)\dfrac{\pi}{2},\forall n\in z\end{cases}$

LEVEL - IV

1. Assertion(A) : $f(x)=\dfrac{\sin\{[x]\pi\}}{1+x^2}$ is continuous on R. where [x] & {x} denotes greatest integer function and fractional part of x.

Reason (R): Every constant function is continuous on R

1) Both A and R are true and R is the correct explanation of A

2) Both A and R are true and R is not the correct explanation of A

3) A is true but R is false

4) R is true but A is false

2. **Assertion(A):** $f(x)=x\left(\frac{1+e^{1/x}}{1-e^{1/x}}\right),\ (x\neq 0),$

f(0) = 0 is continuous at x = 0.

Reason(R) : A function is said to be continuous at 'a' if both limits are exists and equal to f(a).

1) Both A and R are true and R is the correct explanation of A

2) Both A and R are true and R is not the correct explanation of A

3) A is true but R is false

4) R is true but A is false

3. **Match the following :**

List - I	**List - II**
p) The no.of points of discontinuityof the function $f(x)=\frac{1}{\log\lvert x\rvert}$ **is**	**1) 0**
q) If $f(x)=(x+1)^{\operatorname{cosec} x}$ **is continuous at** $x=0$ **then** $f(0)=$	**2) 3**
r) $f(x)=\frac{\cos x-\sin x}{\cos 2x},$	**3) 1**
$x\neq\frac{\pi}{4}$ **If** f **is continuous on**	**4) e**
$\left[0,\frac{\pi}{4}\right]$ **then** $f\left(\frac{\pi}{4}\right)=$	
s) If $f(x)=\frac{\sqrt{1+\sin x}-\sqrt{1-\sin x}}{x}$ **is continuous everywhere then** $f(0)=$	**5)** $\frac{1}{\sqrt{2}}$

Then the correct match for List - I from List - II

	p	q	r	s
1)	1	2	3	4
2)	4	3	1	2
3)	2	4	5	3
4)	2	4	1	5

4. **List - I** **List - II**

List - I	**List - II**
p) f(x) = [x] is continuous on	1) R
q) f(x) = |x| is continuous on	2) R - Z
r) $f(x)=\frac{\lvert x-2\rvert}{x-2}$ is continuous on	3) R – {2}

Then the correct match for List - I from List - II

	p	q	r
1)	1	2	3
2)	1	3	2
3)	2	1	3
4)	2	3	1

5. **Assertion(A) :** $f(x)=\begin{cases} x^2\sin\left(\frac{1}{x}\right), & if\ x\neq 0 \\ 0, & if\ x=0\end{cases}$ **is continuous at x = 0**

Reason(R):

Both $h(x)=x^2,\ g(x)=\begin{cases}\sin\left(\frac{1}{x}\right), & if\ x\neq 0\\ 0, & if\ x=0\end{cases}$

are continuous at x = 0

1) Both A and R are true and R is the correct explanation of A

2) Both A and R are true and R is not the correct explanation of A

3) A is true but R is false

4) R is true but A is false

LEVEL-IV-KEY

1) 1 2) 1 3) 3 4) 3 5) 3

LEVEL-IV-HINTS

1. A is true, R is true and R $\Rightarrow$ A

$\sin n\pi=0,\ for\ n\in z$

2. $\lim_{x\to 0^-} f(x)=\lim_{x\to 0^+} f(x)$

3. Simplify

4. $[x]$ is continuous on R - Z

$\lvert x\rvert$ is continuous on R

5. A is true but R is not correct

because, $\lim_{x\to 0} g(x)=$ doesnot exists

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